Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2014-II-Q04] | Part A (Q1-Q5)

Answer: 447

1. Let the digit integers be $u = 10a + b, v = 100a + 10b + c$. Then $u/99 + v/999 = 33/37$, or $111u + 11v = 9801$.

2. Modulo eleven, u is divisible by eleven, forcing $a = b$. Divide by eleven and substitute $u = 11a, v = 110a + c$ to get $221a + c = 891$.

3. With $0 \leq c \leq 9$, only $a = 4, c = 7$ work, so the answer is 447.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2023-II-Q05] | Part A (Q1-Q5)

Answer: 719

1. Write $r = a/b$ with $\gcd(a, b) = 1$, and put $g = \gcd(55, b) \in \{1, 5, 11, 55\}$. The reduced form of $55r$ is $(55a/g)/(b/g)$.

2. The equal-sum condition becomes $(g - 1)b = (55 - g)a$. The values $g = 1, 55$ are impossible for positive a, b.

3. For $g = 5$, $2b = 25a$, forcing $(a, b) = (2, 25)$. For $g = 11$, $5b = 22a$, forcing $(a, b) = (5, 22)$. Each has the required gcd and indeed equal sums.

4. The sum is $2/25 + 5/22 = 169/550$, which is reduced, and $p + q = 719$.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2006-I-Q06] | Part B (Q6-Q10)

Answer: 360

1. Each equals $(100a + 10b + c)/999$. There are $10 \cdot 9 \cdot 8 = 720$ triples, and each digit appears in each position 72 times.

2. The sum is $72(0 + 1 + \cdots + 9)(100 + 10 + 1)/999 = 72 \cdot 45/9 = 360$.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2003-I-Q14] | Part C (Q11-Q15)

Answer: 127

1. If 251 occurs later, multiply by a suitable power of ten and take the fractional part. The new reduced denominator divides the original one and the decimal starts $0.251\ldots$. Thus it suffices to minimize the denominator of a fraction in $[251/1000, 252/1000)$.

2. Such a fraction exceeds $1/4$. Let $h = 4m - n \geq 1$. The strict upper endpoint gives $h/(4n) < 1/500$, hence $n > 125h \geq 125$.

3. For $n = 126$, the only possible h is 1, but $4m = 127$ is impossible. At $n = 127$, $m = 32$ gives $32/127 \in [0.251, 0.252)$ and is reduced. Therefore the least denominator is 127.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2022-I-Q13] | Part C (Q11-Q15)

Answer: 392

1. These decimals are $k/9999$ for $1 \le k \le 9999$. A positive integer m occurs as a reduced numerator iff $m \le D$ and $\gcd(m, D) = 1$ for some divisor D of $9999 = 9 \cdot 11 \cdot 101$.

2. If m avoids all three primes $3, 11, 101$, choose $D = 9999$, giving $\varphi(9999) = 6000$ numerators. If m is divisible by 3 only, the largest allowed denominator is 1111, giving $\lfloor 1111/3 \rfloor - \lfloor 1111/33 \rfloor - \lfloor 1111/303 \rfloor = 334$.

3. If 11 alone divides m, use $D = 909$, yielding $\lfloor 909/11 \rfloor - \lfloor 909/33 \rfloor = 55$. If both $3, 11$ divide m but not 101, $D = 101$ permits $33, 66, 99$. Any divisibility by 101 forces the maximum coprime denominator below m, so contributes nothing.

4. Thus $N = 6000 + 334 + 55 + 3 = 6392$, with remainder 392.

Source / 来源